

Assignment 3

Please note that this assignment requires a substantial amount of effort, but you are explicitly requested to use GenAI tools (such as Claude, ChatGPT, etc.) as support.

Content-Based Video Retrieval System

The main goal of this assignment is to develop a *content-based video retrieval system* that can be used for finding small video segments of interest (with a duration of a few seconds) in a moderately large dataset of 200 videos (25 hours content). More specifically, we will focus on *Known-Item Search* (KIS) and *Visual Question Answering* (VQA) queries in context of the *Video Browser Showdown* (VBS) – see <https://www.videobrowsershowdown.org>. The system must use a GUI as a frontend and be able to send found segments via the REST API to the *Distributed Retrieval Evaluation Server* (DRES). There is an OpenAPI specification for it and you should use it to generate code for communication with DRES. [See here for more details](#).

V3C Dataset

We will use a small part of the V3C-1 dataset (https://link.springer.com/chapter/10.1007/978-3-030-05710-7_29). More specifically, we will use a subset of about 200 videos, consisting of the video files themselves (with different resolutions) and meta-data (basically upload descriptions from VIMEO used for those videos by the authors). You can download the videos here:

Link: <https://www2.itec.aau.at/owncloud/index.php/s/AcA1pvZlpDrOom5>

Password: IVADL2026

Visual Content Analysis (Backend)

Analyze the content in the video dataset :

1. Extract segments (shots) based on the provided info from TransNet2 (file: scenes_v3c1_200.zip), which provides start and end frame segmentation.
2. Be creative and create an architecture for a video content search system, based on video content analysis; e.g., with
 - a. CLIP, BLIP, or SigLIP
 - b. Vision Transformers
 - c. Convolutional Neural Networks
 - d. Region-based CNNs
 - e. ...
3. Store meta-data and analysis results in an appropriate database and/or indexing system.
4. Transcode videos (e.g., with ffmpeg) to small versions so that they can be used with your GUI for interactive playback/inspection in the competition.

What methods you choose is entirely up to you, but you must do the analysis on your own (with your own scripts and your own databases/indexes). You do not need to *train* neural networks on your own (but you can), you can use existing architectures pretrained on meaningful datasets and just use them for inference with the extracted keyframes/shots (which is fast enough on CPUs too).

Database, Index, GUI

The results of the video analysis (stored in a database and/or index) should be used by your video search system frontend with a graphical user interface (GUI) that integrate features that allow to

1. Interactively search for video content (e.g., keyframes and shots), either by a text-query, content filtering, browsing, similarity search, and/or recommendations.
2. Interactively inspect found items and play the corresponding video content (your system must integrate a video player to check shots or videos).
3. Send a selected item (e.g., keyframe) to the DRES (Distributed Retrieval Evaluation Server) for evaluation (see details below).

Examples of sophisticated video search systems that support such features are summarized in the videos here: <https://videobrowsershowdown.org/vbs2025-systems-videos/>. A more detailed description of VBS systems is available in this publication: <https://link.springer.com/article/10.1007/s00530-023-01143-5>. If you are interested in participating in the VBS (the next one is in January 2027 in Cambodia), you will find [all details here](#).

4. To enable you to test your system with DRES, I will start test tasks in June on the DRES instance running at <https://vbs.videobrowsing.org>, for which you will get a username and password upon request by email to me (of course you can also download and install your own instance of [DRES](#), if you prefer). The DRES server will be used: (1) to present queries in a browser window, and (2) to collect and evaluate your submissions against ground truth.
5. For the evaluation of your system, we will use a live (interactive) in-class evaluation session using the same DRES instance on <https://vbs.videobrowsing.net>. We will do about 10-15 tasks and for each task you will get a maximum of 5 minutes solve time. The faster you find the correct item and the less mistakes you make, the more points you will get and the achieved points will directly impact your grading for the assignment (each group should at least solve one KIS and one VQA task successfully). DRES uses the following scoring scheme: you get 1000 points if you correctly solve a task in the first second, with a linear decrease to 500 points for correctly solving a task in the last second, and a penalty of 100 points for every wrong submission. Therefore, only make submissions to DRES if you are sure that the found segment is in the target segment presented for the task!

Important Notes about DRES

The DRES server is freely available on GitHub (<https://github.com/dres-dev/DRES>) and supports an HTTP-like API that is specified via OpenAPI (<https://github.com/dres-dev/DRES/blob/master/doc/oas-client.json>)¹. To communicate with DRES the first action is a *login* function call with a valid user and password². Next, you should query for actively running evaluation sessions at DRES with *client/evaluation/list* and save the evaluationId of the corresponding session (for your local instance of DRES this will just be the first one, for the instance that will be running at <https://vbs.videobrowsing.org> it will be “IVADL2026”). The most important API function is *submit*, which needs the evaluationId as first parameter, the task name as the second parameter, and an array of (for us just a single) *ApiClientAnswer* items that consist of a text (can be null), the media item (video ID³), the media collection name (“IVADL”), the start time in milliseconds, and the end time in milliseconds (can be the same as start). Please note that for the latter data you need to know the frame-rate of the video (it is advisable to store it in the database).

Submission and Live-Evaluation

You need to submit your implementation project with all sources (backend, frontend), data (models, keyframes, etc.), and a 4-5 pages report (it should include the architecture with justification of different methods and components, a user documentation, and a reflection on the learnings as well the GenAI-usage) via Moodle until the end of the semester (deadline **July 2, 2026**).

The live evaluation competition will be done via Teams on **July 3, 2026 from 09:00-12:00**:

<https://teams.microsoft.com/meet/398243105423722?p=xwN4gmL0hvelX6zcg1>

During this virtual Teams session I will create tasks for the session “IVADL2026” at the DRES instance running at <https://vbs.videobrowsing.org> and you will use the Remote-Viewer of DRES to see when a task starts and how it looks like. You will try to solve the task as fast as possible and submit the correct segment to the DRES instance as fast as possible and DRES will score your submission.

Moreover, I will ask each group/team individually to share their screen and present and demonstrate me their system.

¹ A more convenient view of the OpenAPI specification can also be seen here:

<https://editor.swagger.io/?url=https://raw.githubusercontent.com/dres-dev/DRES/master/doc/oas-client.json>.

² An instance of DRES will be running at <https://vbs.videobrowsing.org/>. Every group will be provided with a username and a password upon request to me by email. However, the effort for a successful communication with DRES is **not to be underestimated** and should be implemented and tested early.

³ Video name without file extension.